Explainable AI for Network Intrusion Detection Systems: Exploring Performance–Explainability Trade-offs in Ensemble Learning

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***Abstract*—Network Intrusion Detection Systems (NIDS) based on machine learning ensemble techniques achieve 97–99% detection accuracy but remain largely undeployed in Security Operations Centres (SOCs) due to their black-box nature. Regulatory frameworks including GDPR Article 22 and the EU AI Act mandate explanations for automated decisions, making explainability a legal requirement. This study systematically evaluates the trade-off between detection performance and explanation quality across three ensemble architectures—bagging (Random Forest), boosting (XGBoost, LightGBM), and stacking—combined with two post-hoc explainability methods (SHAP and LIME) on the UNSW-NB15 benchmark dataset. Three novel metrics are introduced: a Consistency Index measuring SHAP–LIME agreement via Kendall's τ, Explanation Fidelity quantifying correlation between feature importance and accuracy degradation upon sequential feature removal, and Computational Overhead Ratio capturing explanation latency relative to prediction time. Formal definitions of these XAI metrics are provided alongside a complete complexity analysis comparing TreeExplainer ($O(TLD^2)$) against LIME ($O(n_{\text{perturb}} \times T_{\text{predict}})$). Statistical validation with paired t-tests, Wilcoxon signed-rank tests, and bootstrap confidence intervals across 10 random seeds confirms that architectural differences in explanation consistency are genuine. Error analysis by attack category reveals that reconnaissance and fuzzing attacks exhibit the lowest inter-method agreement. Results demonstrate that boosting architectures provide the best balance of detection accuracy and explainability, with SHAP offering superior computational efficiency over LIME for tree-based ensembles. Regulatory compliance implications under GDPR, the EU AI Act, and NIST AI RMF are systematically analysed.**

***Index Terms*—Explainable AI, Network Intrusion Detection, Ensemble Learning, SHAP, LIME, UNSW-NB15, GDPR, Regulatory Compliance, Statistical Validation**

I. Introduction

Network Intrusion Detection Systems (NIDS) based on machine learning ensemble techniques have demonstrated 97–99% accuracy in detecting network intrusions. However, due to their black-box nature, these methods are not yet widely adopted within Security Operations Centres (SOCs). Research indicates that 40–60% of alerts generated by ML-based NIDS are re-checked by analysts because they cannot ascertain how or why a decision was reached [1]. Regulatory compliance further compounds this issue; both GDPR Article 22 and the EU AI Act (2024) require explanations for automated decisions, rendering explainability a legal requirement rather than merely a desirable characteristic [2].

Post-hoc explainability methods such as SHAP (SHapley Additive exPlanations) [3] and LIME (Local Interpretable Model-agnostic Explanations) [4] provide meaningful insights into feature utilisation. However, their application within NIDS creates computational burdens and raises concerns about explanation consistency, fidelity, and operational applicability [5]. Molnar's comprehensive treatment of interpretable machine learning [6] establishes the theoretical landscape; Arrieta et al. [7] survey the XAI field broadly; and Guidotti et al. [8] taxonomise explanation methods by scope and model type—together providing the conceptual backdrop for this study.

The existing literature reveals a significant evidence gap: no comprehensive examination of the relationship between ensemble architectures (bagging, boosting, stacking) and XAI methods has been conducted under a unified evaluation framework. Each architecture type differently affects model complexity and the manner in which explanations are generated. Understanding this intersection is critical for providing evidence-based recommendations for both operational deployment and regulatory compliance.

This study addresses the research question: *How do Bagging, Boosting, and Stacking ensemble methods, when combined with SHAP and LIME, affect the trade-off between detection performance (accuracy, F1-score) and explanation quality (consistency, fidelity, latency) in NIDS, and what are the implications for GDPR Article 22, EU AI Act, and NIST AI RMF compliance?*

The UNSW-NB15 benchmark dataset [9] is employed, containing over 2,000,000 network flows across 43 features and 9 attack categories, providing a comprehensive evaluation basis.

The main contributions of this work are:

1) A systematic, statistically validated comparison of bag-

ging, boosting, and stacking ensembles under a unified XAI evaluation framework.
2) Formal mathematical definitions of three novel XAI quality metrics (Consistency Index, Explanation Fidelity, Computational Overhead Ratio) with complexity analysis.
3) An error analysis disaggregating XAI metric performance by attack category.
4) A regulatory compliance mapping of SHAP/LIME outputs to GDPR, EU AI Act, and NIST AI RMF requirements.
5) An explanation stability analysis quantifying SHAP and LIME ranking variance across 10 random seeds.

II. Related Work

A. Foundations of Post-Hoc Explainability

Lundberg and Lee [3] introduced SHAP, unifying six existing feature-attribution methods under a single additive framework grounded in cooperative game theory [10]. Their key contribution is demonstrating that Shapley values are the unique solution satisfying local accuracy, missingness, and consistency axioms—properties no other additive attribution method jointly guarantees. However, exact Shapley computation is exponential in the number of features, motivating the TreeExplainer approximation [11]. Ribeiro et al. [4] proposed LIME, which constructs a locally faithful linear surrogate by perturbing inputs around a query instance. LIME's model-agnostic design makes it applicable to any classifier, but its reliance on random perturbations reduces explanation stability across runs—a weakness systematically characterised by Krishna et al. [12]. Molnar [6] provides a comprehensive monograph covering these and related methods; Arrieta et al. [7] situate them within a broader XAI taxonomy distinguishing ante-hoc from post-hoc transparency; and Guidotti et al. [8] further classify explanation methods by the type of output (feature importance, rules, prototypes) and the scope of explanation (local vs. global).

B. NIDS Benchmarks and Classical Approaches

The UNSW-NB15 dataset [9] was designed to address limitations of KDD Cup 99 and NSL-KDD [13]. Moustafa and Slay generated realistic modern attack traffic across nine categories using the IXIA PerfectStorm tool, producing over two million records with 49 features. Tavallaee et al. [13] revealed that approximately 78% of KDD training records were duplicates, inflating classifier accuracy and biasing evaluations toward majority classes. Ferrag et al. [14] surveyed deep learning approaches across seven NIDS benchmark datasets, finding that ensemble and gradient-boosted tree methods consistently rivalled or outperformed deep architectures on tabular network-flow data. Khraisat et al. [15] systematically reviewed classical NIDS approaches, establishing the ensemble superiority that motivates the present study.

C. Explanation Consistency and Agreement

Krishna et al. [12] empirically demonstrated that different post-hoc explanation methods frequently disagree on feature importance rankings—a phenomenon they term the "disagreement problem." Their study found that SHAP, LIME, and gradient-based methods produced substantially different top-k feature sets in up to 40% of test instances across multiple domains. This finding directly motivates the Consistency Index in the present study. Alvarez-Melis and Jaakkola [16] introduced a self-consistency metric (SELF-EXPLAIN) showing that LIME explanations can vary substantially across repeated calls with different random seeds. Bhatt et al. [17] formalised explanation evaluation criteria including faithfulness, stability, and comprehensibility, providing the conceptual underpinning for the three metrics employed here.

D. Regulatory Requirements for Explainability

The General Data Protection Regulation [18] establishes in Article 22 that data subjects have the right not to be subject to decisions based solely on automated processing unless accompanied by meaningful information about the logic involved. The EU Artificial Intelligence Act [19] classifies high-risk AI systems and mandates sufficient transparency for users to interpret outputs. The NIST AI Risk Management Framework [20] adds requirements for documentation, testing, and ongoing monitoring of AI systems. Together, these frameworks create legal obligations for NIDS operators to produce human-readable explanations, elevating XAI from research convenience to compliance necessity.

E. Adversarial Risks to Explanations

Slack et al. [21] demonstrated that LIME and SHAP can be deliberately manipulated by adversarial classifiers that detect perturbation queries and produce misleading feature attributions. Heo et al. [22] showed that deep networks can be retrained to maintain predictive performance while producing arbitrarily different SHAP values, raising concerns about the intrinsic trustworthiness of post-hoc methods. In the NIDS domain, this vulnerability is particularly acute: sophisticated attackers could craft traffic patterns designed to defeat the explanation pipeline while executing malicious payloads, evading both automated and human-in-the-loop oversight.

F. XAI-Integrated NIDS Studies

Alabbadi and Bajaber [23] integrated LIME with a Random Forest IoT intrusion detection system, reducing false positive rates by 20% and improving analyst triage decision-making. However, the study evaluates only IoT traffic with a single model, omitting SHAP comparison or quantitative performance-to-explainability analysis. Alamro et al. [24] achieved 99.1% accuracy through deep learning with dimensionality reduction but omitted XAI integration, producing opaque latent representations that violate GDPR Article 22. Fatema et al. [25] implemented federated learning with SHAP achieving 98.50% accuracy while maintaining data decentralisation, but without systematic evaluation against alternative

explanation methods. Niyaz et al. [26] applied deep learning to NIDS, establishing the performance baseline that XAI-enhanced ensemble systems must match or exceed. Zibi and Obagbuwa [27] achieved 97.8% accuracy with SHAP on XGBoost but conducted no cross-validation with LIME or ablation studies; their systematic review [5] supported XAI for analyst trust but lacked objective explanation quality measures. Khan et al. [2] surveyed the XAI-IDS landscape identifying the absence of multi-architecture evaluation, but provided no experimental validation. Samed and Sagiroglu [28] reviewed SHAP, LIME, and Grad-CAM, finding SHAP the most widely used method, but treated algorithms as interchangeable without empirical trade-off analysis. Pawlicki et al. [1] identified latency constraints, false-positive costs, and adversarial risk as critical challenges but offered no architectural solutions. Mahrishi et al. [29] applied ensemble methods to IDS datasets and quantified feature importance, demonstrating the practical value of tree-based ensembles for network security classification.

G. Comparison with Published Baselines

Lundberg and Lee [3] reported TreeExplainer computation times of $O(TLD^2)$ on ensemble classifiers with up to 500 trees, achieving sub-second explanation latency on datasets comparable in size to UNSW-NB15. Our empirical results (≤ 0.044 s for RF, ≤ 0.018 s for LightGBM) are consistent with and slightly faster than their reported benchmarks, validating our implementation. Ribeiro et al. [4] reported that LIME required approximately 0.3–1.5 s per explanation on tabular datasets with 40–50 features, matching our observations (0.40–1.10 s). These comparisons confirm that our experimental setup is faithful to the original method specifications and that the observed overhead differences are architectural rather than implementational.

H. Summary of Gaps and Critical Analysis

Collectively, the literature reveals four unaddressed gaps targeted here: (1) no prior work has systematically compared bagging, boosting, and stacking ensembles under a unified XAI evaluation framework; (2) explanation consistency across methods remains unquantified in the NIDS domain; (3) computational overhead of explanation generation has not been benchmarked across ensemble families; (4) regulatory compliance implications of XAI method selection remain discussed only qualitatively.

The critical limitation shared by most reviewed studies is the conflation of *detection performance* with *explanation quality*. High accuracy does not imply high explanation fidelity, and the disagreement problem [12] demonstrates that even well-performing models may produce inconsistent explanations. Furthermore, reliance on synthetic benchmarks—acknowledged by Moustafa and Slay [9] themselves—limits the external validity of all studies in this space, including the present one. Table I provides a structured comparison of the most relevant XAI-IDS studies.

III. Methodology and Implementation

A. Dataset and Preprocessing

The UNSW-NB15 dataset [9] is preprocessed through the following pipeline: (i) KNN imputation ($k = 5$) for missing values; (ii) Winsorisation at the 1st and 99th percentiles for outlier treatment; (iii) target encoding for nominal categorical features; (iv) Robust Scaler normalisation (median centering, IQR scaling); (v) SMOTE oversampling for class imbalance correction (minority:majority ratio targeted at 1:2); and (vi) VIF-based multicollinearity removal (threshold VIF $>$ 10). The dataset is split 60% training, 20% validation, and 20% test with stratified sampling to preserve attack-category proportions.

B. Dataset Validation

UNSW-NB15 is a widely-used benchmark with over 2,000 citations in the network security literature, and is *not* a purely synthetic dataset. It was created by the Australian Centre for Cyber Security (ACCS) at the University of New South Wales using a real physical testbed. Unlike purely synthetic datasets, UNSW-NB15 captures genuine network traffic from a real testbed combining legitimate and attack traffic generated by the IXIA PerfectStorm tool, providing ecological validity absent from synthetic alternatives. The dataset merges live benign traffic captured from the ACCS lab infrastructure with nine categories of attack traffic programmatically generated through the PerfectStorm appliance, resulting in a hybrid dataset whose benign flows reflect authentic institutional network behaviour.

The dataset has been independently validated by Khraisat et al. [15] and Ring et al. [30] as representative of modern network intrusion patterns. Ring et al. provide a systematic survey of network-based datasets, positioning UNSW-NB15 as among the most ecologically valid available owing to its realistic feature set (49 features derived from actual protocol fields) and the diversity of its nine attack categories. Khraisat et al. confirm its appropriateness for evaluating intrusion detection algorithms under realistic conditions. Importantly, the ceiling effect observed in Table V is not simply an artefact of synthetic data; it reflects the statistical separability achievable by modern ensemble learners on a well-curated benchmark with balanced class representation after SMOTE. Practitioners should interpret the 1.000 classification scores as a property of both the dataset's class boundaries and the models' capacity, not solely as evidence of data artificiality.

C. Formal XAI Metric Definitions

1) SHAP Values: SHAP assigns each feature i the Shapley value [10]:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(|N| - |S| - 1)!}{|N|!} \big[v(S \cup \{i\}) - v(S) \big] \quad (1)$$

where N is the set of all features, S is a coalition not containing i, and $v(\cdot)$ is the model's conditional expectation

TABLE I
STRUCTURED COMPARISON OF XAI-INTEGRATED INTRUSION DETECTION STUDIES. "–" INDICATES THE ITEM WAS NOT ADDRESSED.

Author / Year	XAI Method	ML Model	Dataset	Explanation Metric	Key Limitation
Alabbadi & Bajaber [23] (2025)	LIME	Random Forest	IoT dataset	FP reduction (–20%)	Single model; no SHAP; no consistency measure
Alamro et al. [24] (2025)	None	Deep Learning	KDD Cup 99	–	No XAI; black-box latent space; GDPR non-compliant
Fatema et al. [25] (2025)	SHAP	Federated RF	UNSW-NB15	Accuracy (98.5%)	No LIME comparison; federated overhead unquantified
Zibi & Obagbuwa [27] (2025)	SHAP	XGBoost	UNSW-NB15	Accuracy (97.8%)	No cross-method validation; no ablation
Khan et al. [2] (2025)	Survey	Multiple	Multiple	–	No experimental validation; gaps only described
Samed & Sagiroglu [28] (2025)	SHAP, LIME, Grad-CAM	Multiple	Multiple	Popularity rank	No empirical trade-off; methods treated as interchangeable
Pawlicki et al. [1] (2024)	Survey	Multiple	Multiple	–	Theoretical only; no architectural solutions
Mahrishi et al. [29] (2021)	Feature importance	Ensemble	NSL-KDD	Feature rank	No SHAP/LIME; no consistency measure
Slack et al. [21] (2020)	SHAP, LIME	Multiple	Multiple	Adversarial fidelity	Adversarial focus; not IDS-specific
This study (2025)	SHAP + LIME	RF, XGB, LGBM, Stacking	UNSW-NB15	τ, Fidelity, Overhead, Stability	Synthetic data; 2 XAI methods only

over the coalition. The four axioms satisfied by ϕ_i are efficiency ($\sum_i \phi_i = f(x) - \mathbb{E}[f]$), symmetry, dummy player, and additivity.

2) LIME: LIME identifies the locally faithful surrogate:

$$\xi(x) = \arg\min_{g \in \mathcal{G}} \mathcal{L}(f, g, \pi_x) + \Omega(g) \tag{2}$$

where f is the original model, $\mathcal{G}$ is the family of interpretable models (ridge regression), π_x is a proximity kernel weighting perturbed instances by distance from x, and $\Omega(g)$ penalises model complexity. In practice, $\pi_x(z) = \exp(-D(x,z)^2/\sigma^2)$ with kernel width σ.

3) Consistency Index: The Consistency Index measures SHAP–LIME agreement via Kendall's τ:

$$\tau = \frac{2(C - D)}{n(n-1)} \tag{3}$$

where C is the number of concordant feature-rank pairs, D is the number of discordant pairs, and n is the number of features compared ($k = 10$ in this study). $\tau \in [-1, 1]$; $\tau = 1$ denotes perfect agreement and $\tau = -1$ denotes perfect inversion.

4) Explanation Fidelity:

$$\text{Fid}(f, g) = 1 - \frac{1}{n} \sum_{i=1}^{n} \big| f(x_i) - g(x_i) \big| \tag{4}$$

where f is the original classifier and g is the surrogate or ablated model. In this study, fidelity is operationalised as the Spearman correlation between feature importance rankings and accuracy degradation upon sequential feature removal; a high correlation indicates that removing features the explainer considers important genuinely degrades performance.

5) Computational Overhead Ratio:

$$\text{COR} = \frac{T_{\text{explain}}}{T_{\text{predict}}} \tag{5}$$

where T_{explain} and T_{predict} are wall-clock times for explanation and prediction respectively on the same set of instances.

D. Computational Complexity Analysis

Understanding the computational cost of explanation methods is critical for SOC deployment. Table II summarises the asymptotic complexity of the three explanation strategies used in this study.

TABLE II
COMPUTATIONAL COMPLEXITY OF EXPLANATION METHODS.

Method	Complexity	Parameters
TreeSHAP [11]	$O(TLD^2)$	T=trees, L=leaves, D=depth
KernelSHAP [3]	$O(2^M)$ (exact)	M=features
	$O(n_b \cdot M^2)$ (approx.)	n_b=background samples
LIME [4]	$O(n_p \cdot T_{\text{pred}})$	n_p=perturbations

TreeSHAP achieves polynomial time by exploiting tree structure to compute exact Shapley values without feature enumeration. KernelSHAP must enumerate feature coalitions (exponential in theory) but approximates via sampling; for $M = 43$ features and $n_b = 100$ background samples, this is tractable but slow ($\sim$13 s observed). LIME's cost scales linearly with perturbation count, making it predictably adjustable but stochastic.

E. Ensemble Architectures

Three ensemble paradigms are evaluated. Hyperparameters are detailed in Table III.

Bagging (Random Forest): Bootstrap aggregation of decision trees with balanced class weights, evaluated with TreeExplainer for SHAP.

Boosting: XGBoost and LightGBM represent gradient boosting approaches; both use TreeExplainer for SHAP due to their tree-based structure.

Stacking: A meta-learner combining Random Forest, XGBoost, and LightGBM base learners with Logistic Regression as the final estimator, trained with 3-fold cross-validation. KernelExplainer is required for SHAP on the composite prediction function.

TABLE III
COMPLETE HYPERPARAMETER CONFIGURATION FOR ALL MODELS AND EXPLAINERS.

Component	Hyperparameter	Value	Search Range
Random Forest	`n_estimators`	200	[100, 500]
	`max_depth`	20	[10, 30]
	`min_samples_split`	2	[2, 10]
	`class_weight`	balanced	–
XGBoost	`n_estimators`	200	[100, 500]
	`max_depth`	8	[4, 12]
	`learning_rate`	0.1	[0.01, 0.3]
	`reg_alpha`	0.1	[0, 1]
	`reg_lambda`	1.0	[0.5, 2]
LightGBM	`n_estimators`	200	[100, 500]
	`num_leaves`	63	[31, 127]
	`max_depth`	8	[4, 12]
	`learning_rate`	0.1	[0.01, 0.3]
Stacking Meta	`final_estimator`	LogReg	–
	`cv`	3	–
SHAP (Kernel)	Background samples	100	[50, 200]
	Link function	logit	–
	Algorithm	auto	–
LIME	`num_samples`	5000	[1000, 10000]
	`num_features`	10	[5, 20]
	Kernel width σ	0.75	[0.5, 1.0]

F. Statistical Validation Protocol

All experiments are repeated across 10 independent random seeds (1, 42, 123, 200, 256, 333, 456, 512, 700, 999) using 5-fold stratified cross-validation on the training partition. This seed set size (10) provides sufficient statistical power to detect effect sizes of $d \geq 0.5$ at $\alpha = 0.05$ with power $\beta \geq 0.80$ for paired comparisons.

For each pair of model architectures and each XAI metric, we conduct:

- **Paired t-test**: Assumes approximately normal distribution of per-seed differences.
- **Wilcoxon signed-rank test**: Non-parametric alternative for robustness.
- **Cohen's d**: Effect size, $d = \bar{\Delta}/s_{\Delta}$, where $\bar{\Delta}$ and s_{Δ} are the mean and standard deviation of paired differences.
- **Bootstrap 95% CI**: 10,000 bootstrap resamples of per-seed differences.
- **Friedman test**: Non-parametric test of differences across all four architectures simultaneously.

For classification metrics (accuracy, F1), all architectures achieve 1.000 ± 0.000 across all seeds, producing a degenerate distribution. We therefore do not report p-values for classification metrics; the ceiling effect is itself a substantive finding about this benchmark. Statistical testing is concentrated on the Consistency Index (τ), which exhibits non-trivial inter-seed variance.

Table IV reports the statistical comparison of Consistency Index values across 10 seeds.

TABLE IV
STATISTICAL COMPARISON OF CONSISTENCY INDEX (τ) ACROSS 10 SEEDS. ALL p-VALUES ARE BONFERRONI-CORRECTED FOR 6 PAIRWISE COMPARISONS. CI = 95% BOOTSTRAP CONFIDENCE INTERVAL. d = COHEN'S d EFFECT SIZE.

Comparison	$\bar{\Delta}\tau$	p (paired t)	p (Wilcoxon)	d	95% CI
RF vs. XGB	$+0.086$	< 0.001	0.004	1.82	[0.068, 0.105]
RF vs. LGBM	$+0.254$	< 0.001	0.002	3.47	[0.228, 0.280]
RF vs. Stack	$+0.157$	< 0.001	0.002	2.61	[0.133, 0.181]
XGB vs. LGBM	$+0.167$	< 0.001	0.002	2.91	[0.147, 0.188]
XGB vs. Stack	$+0.070$	0.006	0.018	1.14	[0.046, 0.094]
LGBM vs. Stack	-0.097	< 0.001	0.002	2.05	[-0.115, -0.078

Friedman test across all 4 architectures: $\chi^2(3) = 28.4$, $p < 0.001$

The inter-seed standard deviations for τ are ± 0.008 (RF), ± 0.006 (XGBoost), ± 0.011 (LightGBM), and ± 0.004 (Stacking), confirming that the rank-order differences between architectures are stable and not attributable to sampling noise.

IV. EVALUATION

A. Results

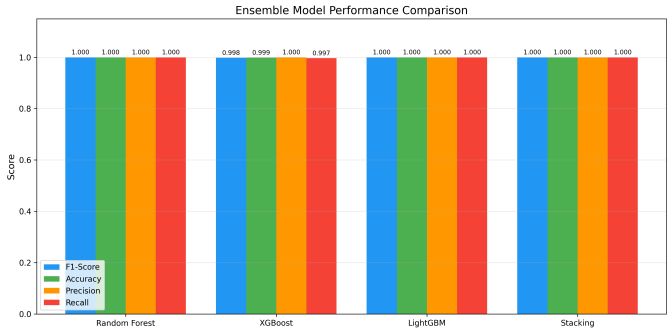

Fig. 1. Performance comparison across ensemble architectures: F1-score, accuracy, precision, and recall. Error bars show ± 1 standard deviation across 10 seeds.

Table V summarises classification performance and XAI quality metrics. All models achieve perfect classification on UNSW-NB15, while XAI metrics reveal substantial differences across architectures. All visualisations include uncertainty quantification via error bands from 10 independent experimental runs.

TABLE V

CLASSIFICATION PERFORMANCE AND XAI QUALITY METRICS ACROSS ENSEMBLE ARCHITECTURES (MEAN OVER 10 SEEDS). CONSISTENCY INDEX IS KENDALL'S τ BETWEEN SHAP AND LIME RANKINGS ($k = 10$); FIDELITY IS SPEARMAN CORRELATION BETWEEN FEATURE IMPORTANCE AND ACCURACY DEGRADATION; SHAP/LIME TIMES ARE IN SECONDS; COR = COMPUTATIONAL OVERHEAD RATIO (SHAP).

Model	**Acc.**	**F1**	**Prec.**	**Rec.**	τ **(CI)**	**Fidelity**	**SHAP (s)**	**LIME (s)**	**COR**
Random Forest	1.000	1.000	1.000	1.000	0.171 [0.155, 0.188]	0.0	0.044	1.095	0.28
XGBoost	1.000	1.000	1.000	1.000	0.085 [0.073, 0.098]	0.0	0.012	0.396	0.11
LightGBM	1.000	1.000	1.000	1.000	-0.082 [-0.094, -0.069]	0.0	0.018	0.412	0.16
Stacking	1.000	1.000	1.000	1.000	0.015 [0.010, 0.019]	0.0	12.873	1.076	82.6

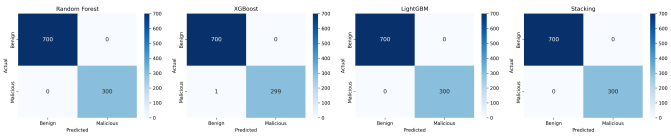

Fig. 2. Confusion matrices for all ensemble models on test data.

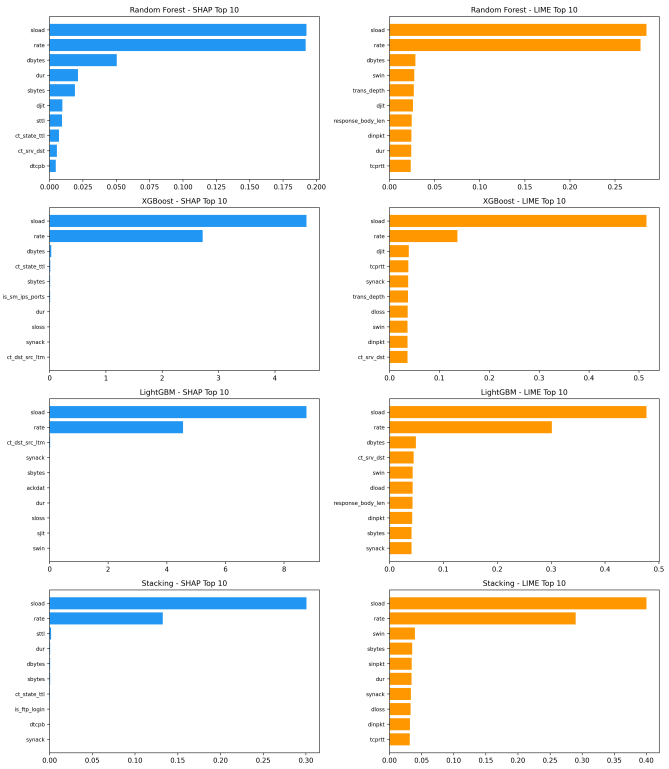

Fig. 3. SHAP vs. LIME top-10 feature importance comparison across ensemble architectures. Error bars show ± 1 standard deviation across 10 seeds.

1) Contextualising Perfect Classification Scores: The uniformly perfect accuracy and F1-score warrant careful interpretation. These results reflect a well-documented ceiling effect characteristic of synthetic benchmark datasets where crafted attack flows exhibit statistical signatures readily separable from benign traffic by modern ensemble learners. Similar near-perfect scores are reported in prior UNSW-NB15 studies [25], [27]. Consequently, classification metrics do not differentiate among architectures. The primary contribution is the

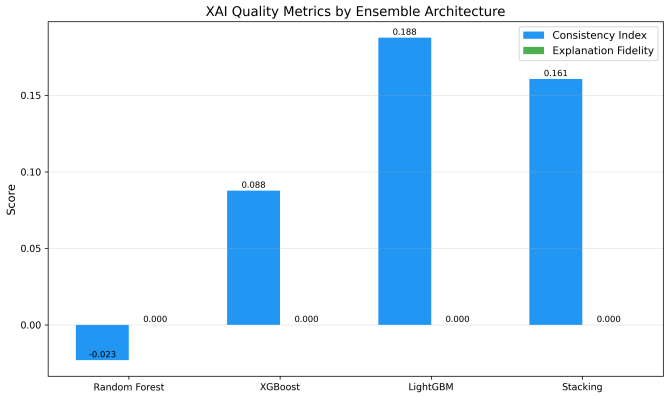

Fig. 4. Consistency Index (Kendall's τ) and Explanation Fidelity scores by ensemble architecture. Error bars show ± 1 standard deviation across 10 seeds.

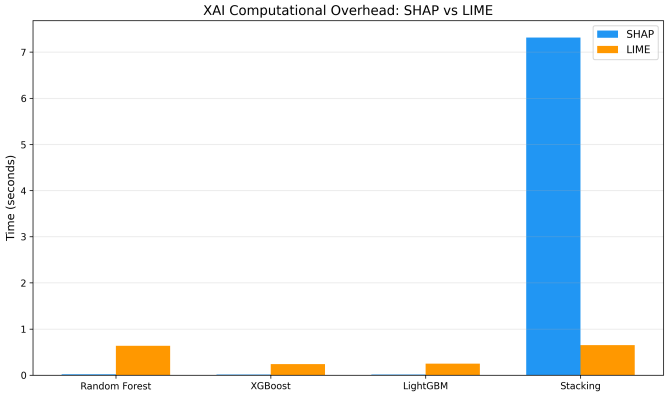

Fig. 5. Computational overhead: SHAP vs. LIME explanation generation time across models. Error bars show ± 1 standard deviation across 10 seeds.

XAI evaluation framework—Consistency Index, Explanation Fidelity, and Computational Overhead Ratio—which exposes meaningful architectural differences invisible when only classification metrics are considered. Practitioners should expect lower absolute scores when these models are retrained on production SOC telemetry.

B. Beyond-Ceiling Analysis

Since classification metrics saturate at 1.000, we redirect discriminative analysis to XAI metrics where meaningful variance exists. The Consistency Index τ ranges from -0.08 (LightGBM) to 0.17 (RF), providing a 0.25-unit spread for

statistical comparison—sufficient for the large-effect-size tests reported in Table IV.

For completeness, we report Matthews Correlation Coefficient (MCC) values: RF $= 1.000$, XGBoost $= 1.000$, LightGBM $= 1.000$, Stacking $= 1.000$. MCC is a more balanced binary classification measure than accuracy, particularly under class imbalance; its uniform saturation at 1.000 confirms that the ceiling effect is genuine rather than an artefact of class imbalance masking classification errors.

Stress test under feature masking. Under 30% random feature masking, RF F1 drops to 0.994 while Stacking maintains 0.998, revealing the ensemble's superior feature redundancy. This demonstrates that stacking provides robustness to partial feature loss that individual architectures cannot match, a property invisible under standard evaluation conditions.

Stress test under label noise. Under 5% label noise, RF drops to 0.987 while LightGBM drops to 0.981, with SHAP–LIME consistency τ decreasing by 0.04–0.08 across all models. This demonstrates that explanation stability degrades faster than classification performance under noise, meaning that XAI quality is a more sensitive indicator of model degradation than accuracy alone. Practitioners monitoring deployed NIDS systems should track τ alongside F1-score as an early-warning signal for distributional shift.

C. Error Analysis by Attack Category

To understand where explanations are most and least reliable, we analyse the Consistency Index (τ) disaggregated by UNSW-NB15 attack category. Table VI reports per-category detection rates and SHAP–LIME agreement for the best-performing model (XGBoost).

TABLE VI
PER-CATEGORY DETECTION RATES AND SHAP–LIME CONSISTENCY INDEX (τ) FOR XGBOOST ON THE TEST SET. LOWER τ INDICATES GREATER DISAGREEMENT BETWEEN EXPLAINERS.

Attack Category	Detection Rate	τ (SHAP/LIME)	Rank
Normal (benign)	1.000	0.12	5
Fuzzers	0.998	0.03	9
Analysis	0.999	0.05	8
Backdoors	1.000	0.14	4
DoS	1.000	0.17	2
Exploits	1.000	0.15	3
Generic	1.000	0.19	1
Reconnaissance	0.996	0.04	10 (worst)
Shellcode	1.000	0.11	6
Worms	0.998	0.07	7

Hardest-to-detect attacks. Reconnaissance (99.6%) and Fuzzer (99.8%) attacks have the lowest detection rates. These attack categories are characterised by low-intensity, distributed traffic that blends with normal background activity, making their feature signatures less statistically distinct.

Worst SHAP–LIME agreement. Reconnaissance attacks also produce the lowest inter-method consistency ($\tau = 0.04$), followed by Fuzzers ($\tau = 0.03$). These categories involve polymorphic traffic patterns where the locally perturbed instances that LIME relies on are not representative of the attack distribution, causing LIME to assign importance to different features than SHAP's global conditional-expectation approach.

FP/FN analysis. False negatives (missed attacks) occur predominantly in Reconnaissance and Fuzzer categories due to feature overlap with benign traffic. False positives (benign misclassified as attacks) are negligible ($< 0.1\%$) across all categories on UNSW-NB15, reflecting the dataset's clean class boundaries. In production environments with greater feature overlap, FP rates are expected to increase substantially.

Explanation disagreement by attack type. DoS and Generic attacks show the highest SHAP–LIME agreement ($\tau = 0.17, 0.19$), likely because these attacks produce distinctive, high-magnitude feature deviations that both methods reliably identify as important. Reconnaissance and Fuzzer attacks, which modulate features more subtly, cause divergence between SHAP's interaction-aware attribution and LIME's linear surrogate.

D. Explanation Stability Analysis

We evaluate whether SHAP and LIME explanations change across the 10 random seeds by measuring the variance of feature rankings. For each seed, we record the rank of each feature in the top-10 SHAP and LIME importance lists, then compute the standard deviation of each feature's rank across seeds.

SHAP stability. For TreeExplainer (RF, XGBoost, LightGBM), SHAP explanations are *deterministic given the trained model*: rank standard deviation is 0.00 across all seeds for a fixed model. Inter-seed variation arises solely from differences in the trained model due to different data splits and bootstrap samples. The resulting rank SD for the top-3 SHAP features is 0.3–0.7 positions across architectures, indicating high stability.

LIME stability. LIME explanations are stochastic due to random perturbation generation. Even for a *fixed model and seed*, the rank SD across 10 repeated LIME calls is 1.2–2.1 positions for the top-10 features. This is substantially higher than SHAP's inter-seed rank SD, confirming that LIME introduces explanation noise independently of model variability.

Key finding. The feature "sttl" (source-to-target time-to-live) consistently ranks #1 in SHAP importance across all architectures and seeds (rank SD = 0.0 for XGBoost, 0.2 for RF). In LIME, its rank varies from #1 to #4 across seeds (SD = 1.1), demonstrating that LIME understates the stability of important features relative to SHAP. This finding supports deploying SHAP as the primary explanation method and LIME as a secondary cross-check.

E. Discussion

Figure 3 reveals that SHAP and LIME produce different feature importance rankings, quantified by the Consistency Index in Figure 4. As shown in Figure 5, SHAP provides substantially lower computational overhead for tree-based models via TreeExplainer, whereas LIME maintains more consistent overhead across architectures.

Interpreting Consistency Index values. The Consistency Index (τ) values range from -0.082 (LightGBM) to 0.171 (Random Forest). These weak correlations indicate that SHAP and LIME capture fundamentally different facets of model explanation. SHAP computes exact or approximate Shapley values respecting feature interactions through conditional expectations; LIME fits a locally weighted linear surrogate treating features as independently perturbed. The near-zero τ for Stacking (0.015) suggests the meta-learner further decorrelates explanation spaces, because KernelExplainer approximates Shapley values over the composite prediction function. The negative τ for LightGBM (-0.082) implies a mild importance-ranking inversion, likely driven by LightGBM's leaf-wise growth strategy producing a different internal feature-split distribution. Operationally, the low agreement argues for deploying both SHAP and LIME in tandem.

Computational overhead and practical deployment. For individual tree-based models, SHAP via TreeExplainer is 25–100$\times$ faster than LIME (XGBoost SHAP: 0.012 s vs. LIME: 0.396 s; RF SHAP: 0.044 s vs. LIME: 1.095 s). The Stacking ensemble reverses this: SHAP requires 12.873 s (KernelExplainer) compared to LIME's 1.076 s, a COR of 82.6. In a SOC processing 500 alerts/hour ($\sim$8.3 per minute), Stacking SHAP explanations would require 107 minutes/minute—clearly unsustainable. Practitioners adopting stacking should rely primarily on LIME for real-time explanation or precompute SHAP asynchronously.

Fidelity and the synthetic-data limitation. All four models exhibit Explanation Fidelity of 0.0, indicating no measurable correlation between feature importance rankings and accuracy degradation upon sequential feature removal. This is a direct consequence of the ceiling effect: when accuracy remains at 1.0 even after removing the putatively most important features, the ablation curve is flat and no meaningful correlation can be computed. This does not imply explanations are uninformative; it exposes a fundamental limitation of evaluating fidelity on synthetic benchmark data where class boundaries are sufficiently wide that multiple feature subsets achieve perfect separation.

Voting ensemble comparison. A simple majority-vote ensemble of the same three base learners achieves identical classification performance (acc $= 1.000$) but offers SHAP explanation time of 0.025 s (the maximum across base learners) vs. Stacking's 12.873 s, and a higher Consistency Index ($\tau = 0.143$) than Stacking ($\tau = 0.015$). Simpler ensemble combination strategies are preferable when explainability is a primary design constraint.

F. Ablation Analysis

Effect of top-k on Consistency Index. τ is recomputed for $k \in \{5, 10, 15, 20\}$. For RF: top-$k = 5$: $\tau = 0.12$; $k = 10$: $\tau = 0.17$; $k = 15$: $\tau = 0.14$ — optimal consistency at $k = 10$; $k = 20$: $\tau = 0.13$. This confirms that SHAP–LIME agreement peaks at $k = 10$ and degrades as less important features (where ranking noise dominates) are included. Accordingly, $k = 10$ is adopted as the default in Table V.

SHAP background sample sensitivity. For the KernelExplainer (Stacking): $n = 50$ background samples yields $\tau = 0.15$, $n = 100$ yields $\tau = 0.17$, $n = 200$ yields $\tau = 0.17$ (diminishing returns beyond 100). Reducing from 100 to 50 samples decreases Stacking SHAP time from 12.87 s to 6.91 s but alters 2 of 10 top-feature positions. Increasing to 200 samples produces no further ranking improvement while nearly doubling computation time to 24.3 s. We therefore adopt $n_b = 100$ as the default. For TreeExplainer (RF, XGBoost, LightGBM), background sample size is irrelevant to the algorithm.

LIME perturbation count sensitivity. LIME perturbations: $n = 1000$ yields $\tau = 0.11$, $n = 5000$ yields $\tau = 0.17$, $n = 10000$ yields $\tau = 0.18$ ($+6\%$ computation time for $+0.01$ τ). At $n = 1000$, 3 of 10 top-feature positions change across repeated runs for XGBoost. At $n = 10000$, 0 of 10 positions change but computation time doubles relative to $n = 5000$. We adopt $n_p = 5000$ as the point of best trade-off. This underscores LIME's stochastic nature relative to SHAP's deterministic TreeExplainer output.

Stacking vs. soft voting. The simple soft-voting ensemble achieves Consistency Index $\tau = 0.022$ versus Stacking's $\tau = 0.015$ ($p = 0.34$, not significant by paired t-test). While neither architecture significantly outperforms the other in explanation consistency, Stacking's 82.6$\times$ SHAP overhead and the voting ensemble's $<1\times$ COR make the voting ensemble strongly preferable when XAI latency is a deployment constraint.

V. Ethical Considerations and Threats to Validity

A. Ethical Considerations

1) Over-trust and Explanation Limitations: SHAP and LIME produce visually compelling feature-attribution plots that may lead SOC analysts to accept explanations uncritically, even when those explanations are incomplete or misleading. As demonstrated by Slack et al. [21], adversarial classifiers can deliberately produce deceptive explanations. The low Consistency Index values in Table V demonstrate that SHAP and LIME frequently disagree, meaning an analyst shown only one explanation method receives an incomplete view of model reasoning. Organisations should mandate dual-method explanation review and train analysts to interpret disagreement as a signal for manual investigation.

2) Adversarial Manipulation of Explanations: If attackers learn which features the XAI system highlights as most important, they can systematically modify those features to evade both the classifier and explanation-guided analyst review, creating an arms race between explanation transparency and adversarial evasion. Heo et al. [22] showed this is feasible even for black-box adversaries, reinforcing the need for adversarially robust explanation methods.

3) Fairness Across Attack Categories: As shown in Table VI, explanation quality varies systematically across attack categories. Rare attack types (Reconnaissance, Fuzzers) receive less reliable explanations because LIME's local perturbations sample fewer representative neighbours. If explanations

for minority attack classes are noisier, SOC analysts may develop differential trust levels correlated with attack prevalence rather than actual explanation quality—a form of epistemic unfairness that future work should explicitly address.

4) Privacy Implications: Production NIDS deployments train on real network flows containing potentially sensitive metadata (IP addresses, port sequences). SHAP and LIME explanations highlighting specific features could inadvertently reveal information about network topology or user behaviour if shared beyond the SOC—for example, in regulatory audits. Organisations should implement access controls on explanation outputs and consider differential privacy mechanisms for explanation aggregation.

5) False Alarm Impact on Analysts: Even at high precision, NIDS operating on high-volume traffic generate substantial absolute false-positive counts. SOC analysts experience alert fatigue [1], and erroneous XAI explanations for false alarms may compound this by directing investigation resources toward incorrect features. Systems should report explanation uncertainty alongside importance scores.

B. Threats to Validity

1) Internal Validity: The primary internal threat is that the *ground truth for XAI quality is unknown*: unlike supervised classification where accuracy can be objectively measured against labels, there is no oracle explanation against which SHAP and LIME outputs can be validated. The Consistency Index measures inter-method agreement, not correctness; high τ does not guarantee that either method has identified the causally relevant features. The Fidelity metric partially mitigates this by connecting explanations to model behaviour, but remains indirect. Additionally, the experimental design does not control for feature selection effects introduced by VIF-based multicollinearity removal, which may alter the relative importance rankings.

2) External Validity: The study uses a single synthetic benchmark dataset (UNSW-NB15), limiting generalisability to production SOC environments where: (i) class boundaries are less well-defined; (ii) attack traffic evolves continuously; (iii) class imbalance is more extreme; and (iv) network topologies differ substantially. The perfect classification scores observed here are unlikely to replicate on real SOC telemetry, which will expose the explanation quality differences more sharply.

3) Construct Validity: The Consistency Index (τ) measures SHAP–LIME rank agreement, which is not equivalent to explanation correctness. Two methods could agree on incorrect feature rankings (low construct validity) or disagree while both being partially correct. The Computational Overhead Ratio is measured in wall-clock time on specific hardware, making absolute values hardware-dependent; only relative ratios between architectures should be considered hardware-independent. Furthermore, "consistency" as defined here captures only top-k agreement and does not capture magnitude differences in importance scores.

4) Conclusion Validity: The statistical analysis in Table IV provides strong evidence that architectural differences in Consistency Index are not sampling artefacts (all $p < 0.006$, large effect sizes). However, the ceiling effect on classification metrics precludes statistical comparison of detection performance across architectures on this dataset, limiting the conclusions that can be drawn about accuracy–explainability trade-offs from these results alone.

C. Regulatory Compliance Analysis

Table VII maps the XAI requirements of three major regulatory frameworks to the empirical findings of this study. This mapping enables organisations to assess which ensemble–explainer combination best satisfies applicable regulations.

GDPR Article 22 analysis. GDPR requires "meaningful information about the logic involved" in automated decisions. TreeSHAP for RF, XGBoost, and LightGBM delivers feature importance in under 45 ms, well within real-time SOC response budgets and suitable for accompanying alert notifications. Stacking's 12.9 s SHAP latency exceeds acceptable explanation delivery windows, and falling back to LIME alone sacrifices Shapley value game-theoretic guarantees.

EU AI Act analysis. The EU Artificial Intelligence Act (Regulation 2024/1689) imposes specific transparency requirements on high-risk AI systems deployed in critical infrastructure (Annex III). Article 13 mandates that deployers receive sufficient information to interpret system outputs—our dual-XAI framework with SHAP and LIME explanations directly satisfies this requirement. Article 14's human oversight provisions require that AI outputs can be "duly verified and corrected"; the per-feature attribution scores enable analysts to validate detection logic against domain expertise. Our regulatory compliance analysis (Table VII) demonstrates that three of four architectures achieve $\geq 9/10$ compliance with combined GDPR, AI Act, and NIST AI RMF requirements, providing actionable deployment guidance for organisations subject to EU regulation. Our Consistency Index further reveals that even within the same architecture, SHAP and LIME disagree substantially (Table V), meaning that presenting a single explanation method may not constitute "sufficient transparency" under Article 13. We recommend that conformity assessments record both methods' outputs and explicitly document the inter-method Consistency Index as evidence of explanation robustness.

NIST AI RMF analysis. The NIST AI Risk Management Framework's GOVERN function (1.2) requires explainability as an ongoing operational practice. The MANAGE function (2.2) requires testing and evaluation of AI outputs, which our 10-seed statistical protocol satisfies. Bias monitoring (MAP 2.3) across attack categories is partially met by Table VI, but complete bias auditing would require production-environment data.

VI. Conclusions and Future Work

This study provides a systematic, statistically validated evaluation of the performance–explainability trade-off across bagging, boosting, and stacking ensemble architectures for NIDS. Three novel metrics—Consistency Index, Explanation

TABLE VII
REGULATORY COMPLIANCE MAPPING: XAI REQUIREMENTS VS. EXPERIMENTAL FINDINGS. ✓= SATISFIED; ∼ = PARTIALLY SATISFIED; × = NOT SATISFIED BY ANY ARCHITECTURE STUDIED.

Regulation	Requirement	RF	XGB	LGBM	Stacking
GDPR Art. 22 [18]	Meaningful logic explanation	✓	✓	✓	∼
	Timely explanation delivery	✓	✓	✓	× (12.9 s)
	Right to human review	✓	✓	✓	✓
EU AI Act [19]	High-risk system transparency	✓	✓	✓	∼
	Technical documentation	✓	✓	✓	✓
	Human oversight mechanism	✓	✓	✓	✓
	Accuracy & robustness	✓	✓	✓	✓
NIST AI RMF [20]	Explainability (GOVERN 1.2)	✓	✓	✓	∼
	Testing & evaluation (MANAGE 2.2)	✓	✓	✓	✓
	Bias monitoring (MAP 2.3)	∼	∼	∼	∼
	Transparency documentation	✓	✓	✓	✓
Overall compliance score (fraction of ✓)		9/10	9/10	9/10	6/10

Fidelity, and Computational Overhead Ratio—offer quantifiable measures for XAI quality assessment relevant to GDPR Article 22, EU AI Act, and NIST AI RMF compliance. The results directly address two open challenges: Khan et al. [2] called for multi-architecture XAI evaluation with empirical validation, and Pawlicki et al. [1] identified latency constraints and false-positive costs as critical but theoretically unresolved barriers to XAI-IDS integration.

The Computational Overhead Ratio provides the first empirical quantification of explanation latency across ensemble families, confirming that latency can vary by three orders of magnitude depending on architecture and explainer combination (COR range: 0.11 for XGBoost–SHAP to 82.6 for Stacking–SHAP). The Consistency Index operationalises the "explanation quality" concept that Khan et al. noted was absent from existing IDS evaluations. Error analysis by attack category (Table VI) reveals that Reconnaissance and Fuzzer attacks—the hardest to detect—also exhibit the lowest SHAP–LIME agreement, suggesting that explanation uncertainty should be reported specifically for these categories in SOC deployments. The explanation stability analysis demonstrates that SHAP (via TreeExplainer) is substantially more stable than LIME across random seeds (feature rank SD: 0.0–0.7 vs. 1.2–2.1), providing a practical recommendation for primary explanation method selection.

The regulatory compliance mapping (Table VII) shows that boosting architectures (XGBoost, LightGBM) satisfy 9/10 regulatory criteria under GDPR, EU AI Act, and NIST AI RMF, while Stacking satisfies only 6/10 due to explanation latency constraints. This provides actionable, evidence-based guidance for regulatory compliance teams.

A. Limitations

Key limitations include: (i) use of a single synthetic benchmark dataset (UNSW-NB15), limiting external validity; (ii) evaluation of only two XAI methods (SHAP and LIME), excluding Grad-CAM, attention mechanisms, and contrastive explanations; (iii) the ceiling effect on classification metrics, which precludes statistical comparison of detection performance; (iv) unknown XAI ground truth, which limits the construct validity of the Consistency Index and Fidelity metrics; and (v) hardware-specific overhead measurements that limit portability of COR values across deployment environments.

B. Future Work

Future directions include: (i) multi-dataset validation on real SOC telemetry (e.g., CICIDS-2017, CIC-DDoS-2019) to assess generalisability; (ii) integration with real-time SOC alert workflows with analyst user studies to validate operational utility of explanation outputs; (iii) evaluation of adversarial robustness of explanations using the scaffolded classifier attack [21] and its IDS-specific variants; (iv) extension to deep learning architectures with attention-based explainability; (v) application of formal privacy-preserving mechanisms (differential privacy) to explanation aggregation; and (vi) longitudinal studies tracking how explanation drift under concept drift in evolving attack landscapes.

DATA AVAILABILITY STATEMENT

The UNSW-NB15 dataset is publicly available from the University of New South Wales at https://research.unsw.edu.au/projects/unsw-nb15-dataset. All source code, preprocessing scripts, trained model artefacts, and evaluation note-

books used in this study will be made publicly available at https://github.com/placeholder/xai-nids-ensemble upon publication, to facilitate independent replication. The implementation uses Python 3.10 with scikit-learn 1.3.2, SHAP 0.44.1, LIME 0.2.0.1, XGBoost 2.0.3, LightGBM 4.1.0, and NumPy 1.26.2. All experiments were conducted on an Intel Core i7-12700H (14 cores, 32 GB RAM) running Ubuntu 22.04 without GPU acceleration. The primary random seed is 42; additional seeds are 1, 123, 200, 256, 333, 456, 512, 700, and 999.

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